

Information Security

Spring Semester 2021 ([252-0211-00L](#))

Overview

Lecturers:

Part 1: [Prof. Dr. Srdjan Capkun](#)

Part 2: [Prof. Dr. David Basin](#)

Assistants:

Part 1: [Dr. Kari Kostiaainen](#), [Dr. Marc Röschlin](#), [Karl Wüst](#), [Daniele Lain](#)

Part 2: [Dr. Dennis Jackson](#), [Dr. Srdan Krstic](#), [Karel Kubicek](#)

Course material:

Published through [Moodle](#)  which also includes a discussion forum.

Lectures:

Thursday 14-16 in Zoom

Friday 14-16 in Zoom

Course introduction will be provided in Zoom on **Thursday February 25**.

Lectures will be recorded videos that are uploaded to Moodle.

For each week of lectures, there will be one live Q&A session in Zoom on Fridays.

The first lecture Q&A is on **Friday February 26**.

Zoom links can be found from [Moodle](#).

Exercises:

Wednesday 16-19 in Zoom

Exercises are uploaded Moodle.

For each exercise topic there will be an exercise session in Zoom on Wednesdays.

The first session is on **Wednesday March 3**.

The exercise sessions will be recorded and uploaded to Moodle.

Zoom links can be found from [Moodle](#).

Credits: 8 ECTS (4V + 3U)

Requirements: None

Language: English

Description

This course provides an introduction to Information Security. The focus is on fundamental concepts and models, basic cryptography, protocols and system security, and privacy and data protection. While the emphasis is on foundations, case studies will be given that examine different realizations of these ideas in practice.

Exercise info

Part I

- The exercise questions will be published approximately one week before each exercise session.
- Students are encouraged to work on the exercises and **attempt to solve the questions first on their own**.
- Master solutions will published a few days later.

- Students are encouraged to **study the master solutions before the exercise session**.
- In the exercise session, the TAs will explain the exercises, discuss alternative solutions, and answer the questions of the students.
- In the exercise session, it will be assumed that the students are already familiar with the questions and the master solutions.
- The exercise sessions are the **primary way of asking questions**. If a student cannot attend the exercise session, or something is not clear after the exercise session, questions can also be asked in the Moodle discussion forum.
- In the discussion forum, **students are encouraged to answer the questions of other students**. TAs will confirm correct answers and reply to any unanswered questions.
- Exercises are not graded, but **working on them** and **attending the exercise sessions** is **highly recommended**.
- The exam questions will be similar to the exercise questions. Exam questions from previous years will be part of exercises.

Part II

As in Part I, however, master solutions will not be published. Instead, you can hand in your solutions to the exercises to receive personal feedback from the tutors. Solutions should be submitted by email to all tutors. Solutions must be received by 23:59 on the Monday after the exercise is published, in order to receive feedback. During the exercise session, the course assistants will explain the correct solutions and answer any questions regarding the exercises. Students who did not submit their solutions are still welcome to attend the session.

Resources

Literature

- Alfred J. Menezes, Paul C. van Oorschot, and Scott A. Vanstone: *Handbook of Applied Cryptography*, CRC Press, 1996 (available online).
- Dieter Gollmann: *Computer Security*, Wiley, 2000.
- Matt Bishop: *Computer Security: Art and Science*, Addison-Wesley, 2002 (available online for ETH members).
- Jonathan Katz and Yehuda Lindell, *Introduction to Modern Cryptography*, Chapman & Hall, 2008
- Charlie Kaufman, Rhadia Perlman, and Mike Speciner, *Network Security: Private Communication in a Public World*, 2nd Edition, 2002.
- William Stallings: *Cryptography and Network Security: Principles and Practice*, 3rd Edition, Prentice Hall, 2002.
- William Stallings: *Network Security Essentials: Applications and Standards*, 2nd Edition, Pearson, 2003.
- Ken Thompson: *Reflections on trusting trust* (available online).
- Wenbo Mao: *Modern Cryptography: Theory & Practice*, Prentice Hall, 2004.